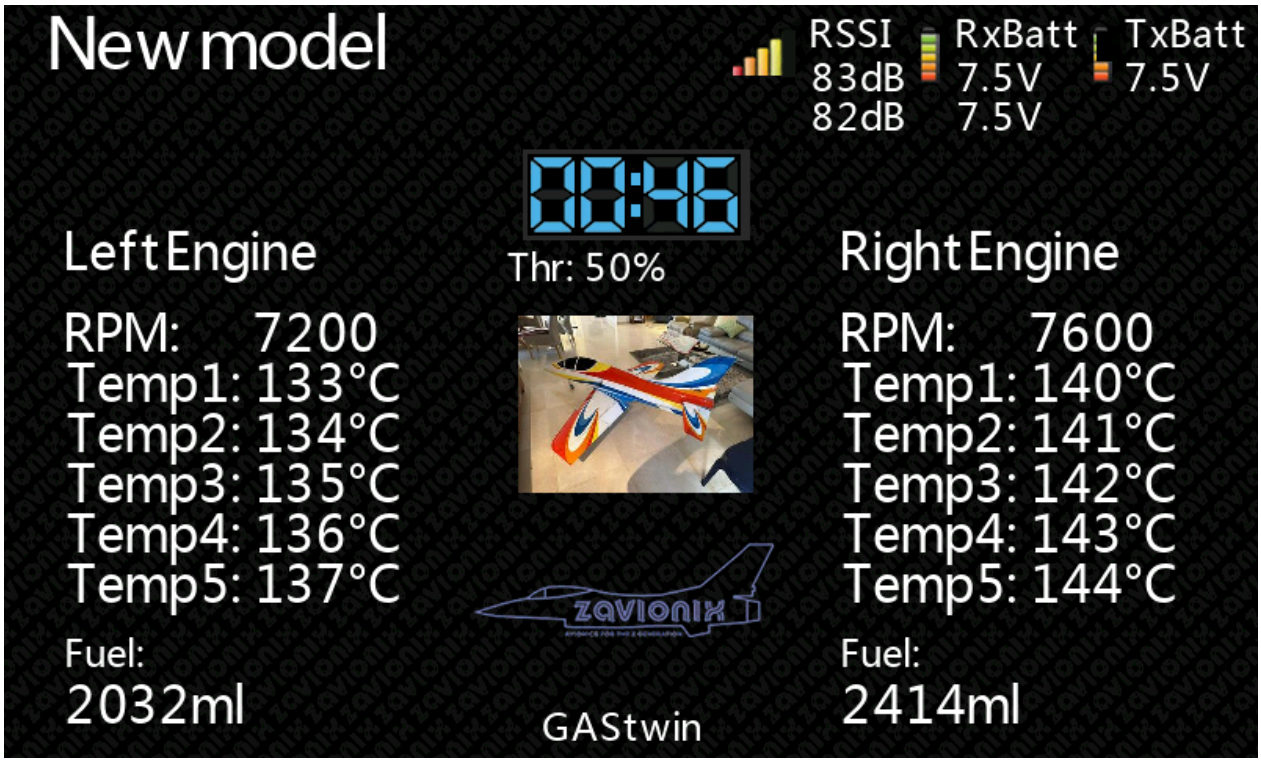
ZAVIONIX / ETHOS

GAScombined

User Manual

Widget V3.2.0 | English | 6 November 2026

Single- and twin-engine gasoline telemetry in one widget. Choose GAS for one engine or GAStwin for two independent engine and fuel displays.



GAStwin in the ETHOS X20S 26.1.3 simulator. All readings are simulated; the model image and numbers are demonstration data.

Based on the GAS user manual, adapted to GAScombined configuration and operation. Both displays retain their own settings when switching views or restarting ETHOS.

Contents: 1 Installation • 2 Display selection and auto-populate • 3 GAS settings • 4 GAStwin settings • 5 Fuel calibration and alerts • 6 Checks and troubleshooting

1. Install and add the widget

Download the Zavionix repository using Code > Download ZIP at https://github.com/i3dm/Zavionix. Extract the archive and locate V3 Ethos Widgets/GAScombined.

1. Back up the model and any existing GAScombined installation.

2. Copy the complete GAScombined folder into the radio storage scripts directory. The published compiled package must contain /scripts/GAScombined/main.luac, its companion modules, images, assets and LowFul.wav. Avoid an extra nested GAScombined folder.

3. Restart ETHOS or reload Lua using the option available in your firmware. When upgrading, install a complete matching package.

4. Open Configure screens (DISP), select or add a screen, choose a full-screen layout, and assign Zavionix GAScombined to the widget area.

5. Open the widget configuration. Choose Widget Display, configure its sensors and tank parameters, then return to the dashboard. Reopen configuration and check that the settings persist.

If installing from the standalone source repository instead, copy the complete src/GAScombined folder; that version starts from main.lua. Simulator fixtures and test radio folders are not installation packages.

GAScombined does not import settings from standalone GAS. Configure it once for your model. GAS and GAStwin also have independent assignments, including their receiver sources and Stick Mode.

Read the dashboard

RPM is direct revolutions per minute, without the turbine widgets' multiplier. Temperature sources must supply numeric Celsius readings. Assign each of the five probes to the intended component; the probe number does not automatically identify a cylinder.

Both displays use ETHOS Timer 1. Configure its start, stop and reset conditions in the model timer settings. The widget displays the timer; it does not configure those controls. Model name and picture come from the ETHOS model. TxBatt comes from the radio; receiver RSSI and battery voltages use your assignments. Throttle follows the transmitter's throttle-stick mapping.

2. Choose a display and populate sensors



GAScombined V3.2.0 configuration: display selection and the auto-populate button. Scroll for the remaining fields.

Widget Display selects GAS or GAStwin. Changing it rebuilds the configuration form for that display. Both sets of settings are retained, but only the selected display runs its fuel alerts.

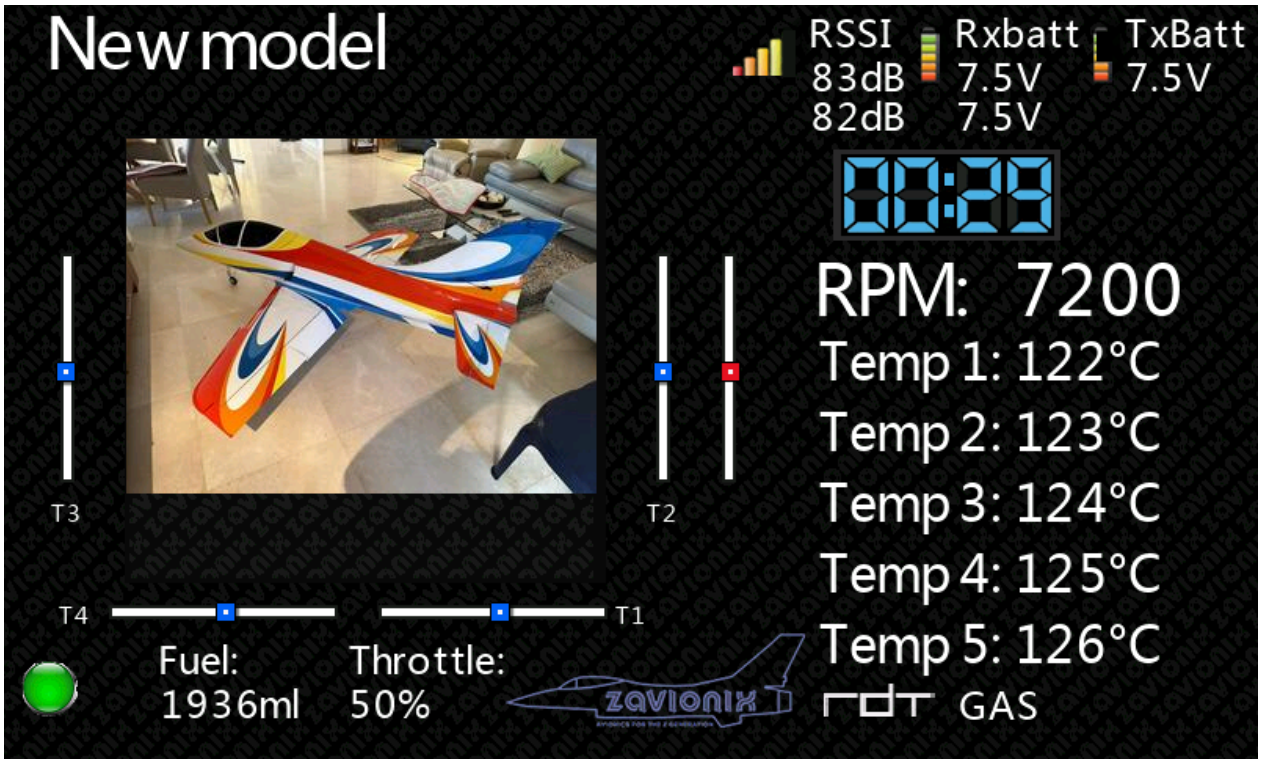
Discover the model's sensors in ETHOS telemetry first. Select the intended display, then press Auto populate menu on the Auto populate sensors row. The button fills recognized assignments for the selected display. Recognized matches may replace existing assignments; unmatched assignments, tank parameters and the other display are preserved. Review every result.

GAS names: RPM; Temp1 through Temp5 (also Tmp1, T1 or Temp 1); FuelUsed, Fuel Used, Consumption or Fuel. A sensor called Fuel must supply consumed ml, not remaining fuel or percent.

GAStwin names must identify the side: RPM_L / RPM_R, RPM L / RPM R, L RPM / R RPM, Left RPM / Right RPM or RPM Left / RPM Right. The same patterns apply to temperatures and fuel, such as Temp1_L and FuelUsed_R. An engine source without a side is not assigned to both engines.

Receiver aliases include RSSI1 / RSSI2, RSSI 1 / RSSI 2, RSSI 2.4G / RSSI 900M and RxBatt1 / RxBatt2. RSSI and RxBatt are single-receiver fallbacks. Auto-populate avoids duplicating one detected source into both receiver entries. Assign any unmatched sources manually.

3. GAS: single-engine settings



The GAS display within GAScombined, with simulated telemetry. It has its own settings, separate from GAStwin and standalone GAS.

Setting	Meaning / available values
Stick Mode	Mode 1, 2, 3 or 4; select the transmitter's stick mode.
Fuel Factor (%)	Consumption multiplier: 1–10000%. 100 means no correction.
Fuel Tank Size (ml)	Full usable tank volume: 1–10000 ml.
Fuel Alert	Enable or disable the low-fuel audio/haptic warning.
Fuel Alert (ml)	Remaining-fuel reserve threshold: 1–10000 ml.
Fuel Alert Rate (s)	Low-fuel warning repeat interval: 1–10 seconds.

Source selectors: RPM Sensor; Temp 1 Sensor through Temp 5 Sensor; Fuel Sensor (cumulative consumed ml); RSSI 1 Sensor and RSSI 2 Sensor; Rxbatt 1 Sensor and Rxbatt 2 Sensor. Keep receiver 1 and 2 assignments distinct. Leave an unavailable source unassigned.

4. GAStwin: two engines, two tanks

The left and right dashboards show independent RPM, five temperatures and remaining fuel. Receiver telemetry, transmitter battery, throttle, Timer 1 and the model image are shared within this display.

Configure the following fields separately for Left and Right. These are the exact label patterns: replace [Side] with Left or Right. Initial values are software defaults, not recommendations for your model.

Setting	Meaning / range / initial value
Stick Mode	One shared setting for GAStwin: Mode 1–4; initially Mode 2.
[Side] Fuel Factor (%)	Consumption calibration: 1–10000%; initially 100%.
[Side] Tank Size (ml)	Full usable volume for that engine: 1–10000 ml; initially 1000 ml.
[Side] Fuel Alert	Enable that tank’s audio/haptic alert; initially enabled.
[Side] Fuel Alert (ml)	Remaining-fuel threshold: 1–10000 ml; initially 300 ml.
[Side] Alert Rate (s)	Independent warning interval: 1–10 s; initially 5 s.
[Side] RPM Sensor	Direct RPM for this engine.
[Side] Temp 1 Sensor through [Side] Temp 5 Sensor	Five independent Celsius sources for this engine.
[Side] Fuel Used Sensor	Cumulative consumed ml for this engine’s tank.
RSSI 1 Sensor / RSSI 2 Sensor	Shared receiver signal-strength sources.
Rxbatt 1 Sensor / Rxbatt 2 Sensor	Shared receiver supply-voltage sources, in volts.

Check each engine separately: vary one source and confirm that only the intended side changes. Assigning one consumption counter to both tanks makes both estimates follow that same counter; it does not measure independent fuel use.

GAStwin shows -- for unavailable numeric readings. Remaining fuel below the configured threshold appears red, even when its audible alert is disabled. Negative fuel remains visible after estimated empty; it signals that calculated consumption has exceeded the configured tank volume.

5. Fuel setup, calibration and alerts

Fuel sources must report cumulative consumed volume in milliliters. Remaining volume, percentage, fuel flow and consumed volume are different quantities. Confirm the sensor’s unit and behavior on the ETHOS telemetry page before assigning it.

Remaining ml = tank size − consumed ml × Fuel Factor / 100

Single-engine example: 1000 ml tank, factor 100 and 850 ml consumed gives 150 ml remaining. With a 300 ml threshold this demonstrates a low-fuel condition. These numbers are examples only.

Twin example: left tank 2400 ml, factor 105 and 350 ml consumed gives 2032.5 ml remaining. Right tank 2900 ml, factor 108 and 450 ml consumed gives 2414 ml. The display uses whole ml, so fractional results are rounded for display.

Set the usable tank volume and begin with the sensor manufacturer’s calibration. After a measured run, compare actual volume used with the widget’s factor-adjusted consumption. For a linear correction: new factor = old factor × actual used / currently calculated used. Equivalently, new factor = 100 × actual used / raw sensor consumed ml. Do not divide by zero. Repeat the measurement, separately for each tank in GAStwin.

After refuelling, reset the upstream consumed-fuel counter through the sensor or the model’s ETHOS reset arrangement. This widget has no Fuel Reset Switch field. Changing tank size, restarting the widget or switching displays does not establish that the consumption counter has reset. Verify the initial consumed reading before each use.

Choose a reserve appropriate to the model, enable the required alert and set its repeat interval. Test the warning on the physical transmitter with a controlled value. GAStwin maintains separate warning intervals for each tank; simultaneous due warnings share one sound. An unavailable GAStwin fuel reading does not generate a low-fuel warning.

Only the currently selected display runs its alerts. Switching to GAS does not continue GAStwin tank warnings in the background. Configure any additional model-wide telemetry alarms in ETHOS as needed for your model.

6. Verification and troubleshooting

Symptom	What to check
Widget is missing from the list	Check scripts/GAScombined, the entry file and all companion files. Restart ETHOS and use a compatible complete package.
Missing reading or --	Check discovery, source assignment and the live source value in ETHOS telemetry. Confirm the selected display's settings.
Wrong engine changes	Check left/right sensor names and assignments; test one engine/probe at a time.
Fuel estimate is incorrect	Check consumed ml versus remaining ml or percent, tank volume, calibration and counter reset.
Negative fuel	Calculated use exceeds configured volume. Check actual fuel, counter reset, capacity and calibration.
No audible fuel warning	Check selected display, alert enable, threshold, repeat interval, live fuel source, radio audio and LowFul.wav.
Timer or image is wrong	Check ETHOS Timer 1 and the model picture settings.
Settings appear different after changing view	GAS and GAStwin retain separate settings. Return to the intended display and inspect its assignments.

Acceptance check: compare all readings with ETHOS telemetry; vary each engine independently; test fuel reset and low-fuel behavior; reopen configuration; then restart ETHOS and confirm both displays retain their assignments and tank parameters. Record the final setup with your model backup.

Edition and screenshot record

Manual edition: 6 November 2026. Covers GAScombined V3.2.0. Adapted from the GAS V3.1.12 manual and checked against GAScombined's configuration, fuel calculation, display-selection and auto-populate implementation.

Screenshots are existing 19 October 2026 captures from the real headless X20S ETHOS 26.1.3 simulator, using simulated DIY telemetry sources. They illustrate the widget and configuration; they are not flight measurements. Physical telemetry accuracy and audible playback were not established by these captures. Other screen sizes may arrange content differently.

Downloads: https://github.com/i3dm/Zavionix Source and updates: https://github.com/i3dm/GAScombined